

Ultimate Demonstration Video Script & Guide

Project: Quantifying Customer Dissatisfaction – A Hybrid NLP Approach

Target Duration: 20 Minutes

This document provides a highly comprehensive script for your 20-minute research project demonstration, with an expanded focus on the Streamlit dashboard functionality.

Video Timeline Overview

Section	Duration	Key Focus
1. Introduction & Research Context	4 Minutes	Problem Statement, Objectives, and Repo Overview
2. Code & Logic Walkthrough	6 Minutes	Phase I (Preprocessing), Phase II (NLP Engines), src/ folder
3. Deep-Dive Dashboard Demo	10 Minutes	Comprehensive walkthrough of all 7 pages & features
4. Conclusion & Research Findings	3 Minutes	Key Insights, Contributions, and Future Work

Part 1: Introduction & Research Context (0:00 – 4:00)

 **Action: Show your GitHub Repository / Project Folder**

Say this:

"Hello everyone. My name is **[Your Name]**, and today I am presenting my final year research project:

1.1 The Problem Statement

In today's e-commerce landscape, customer feedback is the most valuable asset a company has. However, most businesses still rely on binary sentiment analysis—simply labeling a review as 'positive' or 'negative'. This is fundamentally flawed because:

- **Star ratings are misleading:** A customer might give 4 stars but express a specific complaint in the text that gets ignored.
- **Negativity is nuanced:** There is a huge difference between a customer who is 'mildly annoyed' and one who is 'extremely angry'.
- **Root causes are hidden:** Knowing a customer is unhappy doesn't tell a business *why*. Is it the fit? the fabric? or the delivery?

1.2 Research Objectives

My project addresses these gaps with four primary objectives:

1. To develop a **continuous 0–100 Dissatisfaction Index** based on the VADER lexicon.
 2. To implement **Unsupervised Topic Modeling (LDA)** to automatically categorize complaints into business domains.
 3. To build a **Sarcasm Detection layer** using Transformer models (RoBERTa) to catch negative reviews hidden behind positive words.
 4. To integrate these into a **Real-time Analytics Dashboard** for business stakeholders.
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Part 2: Code & Logic Walkthrough (4:00 – 10:00)



Action: Open VS Code and show the `src/` folder

"Let's dive into the technical implementation. My system follows a **3-Phase Architecture**."

Phase I: Preprocessing (`src/preprocessing.py`)

- **Function: `clean_text()` (Lines 66–84):** Note **Line 82** where I use regex `[^a-z\s]` to keep only letters.
- **Function: `replace_slang()` (Lines 59–63):** Converts terms like 'tts' to 'true to size' (Lines 25–40).
- **Function: `lemmatize_text()` (Lines 87–98):** We use POS-aware lemmatization. On **Line 46**, I explicitly **preserve negation words** like 'not' and 'never'.

Phase II: The 3 NLP Engines

1. `sentiment_vader.py`:

- `analyze_single()` (Lines 44–60): Computes the **Dissatisfaction Score** on Line 52:
`round(max(0.0, -compound) * 100, 2) .`

2. `topic_modeling.py`:

- `train_lda_model()` (Lines 66–81): Trains a Gensim LDA model with `num_topics=6`.
- `get_coherence_score()` (Lines 84–91): Calculates the **C_v score** for mathematical proof of topic quality.

3. `sarcasm_detector.py`:

- `_lexical_sarcasm_boost()` (Lines 44–88): Detects 'sarcasm patterns' like quoted praise.
 - `detect_single()` (Lines 146–191): Combines RoBERTa with the lexical layer for the final `irony_prob`.
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Part 3: Deep-Dive Dashboard Demo (10:00 – 20:00)

 **Action: Open the browser at `http://localhost:8501`**

3.1 Page 1: Home & Overview (1.5 mins)

• Feature Walkthrough:

- "The **Overview Metric Cards** give an immediate snapshot: 23,486 reviews and 3 NLP engines."
- "The **Methodology Tab** is crucial. It explains that VADER measures *Intensity*, LDA measures *Theme*, and RoBERTa measures *Nuance*."
- "The **Research Phases Tab** shows our visual pipeline: Data Hub → Hybrid Engine → Analytics Dashboard."
- "This page acts as the 'Control Center' for the entire demonstration."

3.2 Page 2: Data Hub (2 mins)

• Feature Walkthrough:

- "We have three input modes. I've designed the **Upload CSV/Excel** feature (Lines 347–362) so businesses can analyze their own data instantly."
- **Live Analysis Interaction:** "Let's use the **Live Review Text** mode. I'll enter: 'Wow, what a "high quality" dress. It literally fell apart the first time I wore it.'"
- "Note the **RoBERTa Status Indicator**. I'll click 'Load RoBERTa' (Line 406). This downloads the 500MB transformer model. Once loaded, click 'Analyse'."

- **Explaining Results:** "Look at the **Sarcasm Correction** (Lines 617–627). VADER was fooled by 'high quality' and predicted 'Positive'. But RoBERTa identified a 92% irony probability, and my system **overrode the sentiment to Negative (Sarcasm 🤖👉👎)**. This is the hybrid system's greatest strength."

3.3 Page 3: 🛠️ Phase I – Preprocessing (1 min)

- **Feature Walkthrough:**
 - "Click 'Run Preprocessing' (Line 717). This cleans the 23k reviews in real-time."
 - "The **Cleaned Data Tab** shows the raw text next to the lemmatized `processed_text`."
 - "The **Feature Stats Tab** shows box plots (Lines 752–756) for word counts. Angry customers write longer reviews—this chart confirms that theory."
 - "The **Slang Examples Tab** lists all the fashion-domain abbreviations we normalize."

3.4 Page 4: 📊 Phase II – Sentiment (VADER) (1.5 mins)

- **Feature Walkthrough:**
 - "Run the VADER Analysis (Line 821). The **Dissatisfaction Index Gauge** (Lines 845–847) shows our brand health score."
 - "The **Scatter Plot** (Lines 858–859) is vital. It plots VADER Compound against Star Rating. The clear diagonal trend validates that our machine score aligns with human judgment."
 - "The **Review Explorer** (Lines 876–884) allows us to filter by 'Severely Dissatisfied'. We can read the exact texts that scored 90/100."

3.5 Page 5: 🗺️ Phase II – Topic Modeling (LDA) (1.5 mins)

- **Feature Walkthrough:**
 - "Here we train our **LDA Model**. I've added sliders (Lines 913–914) to adjust the number of topics and training passes."
 - "The **Topic Keyword Cards** (Lines 950–958) show the probability weights for words like 'fit', 'fabric', and 'delivery'."
 - "The **Heatmap** (Lines 960–969) is the most actionable chart. Dark red cells identify the specific intersections—like 'Product Quality' at 1-star—that require immediate management attention."

3.6 Page 6: 🤖👉👎 Phase II – Sarcasm (RoBERTa) (1 min)

- **Feature Walkthrough:**
 - "We can analyze a sample of up to 3,000 reviews for sarcasm (Lines 1022–1024)."
 - "The **Donut Chart** (Lines 1086–1087) reveals the sarcasm rate in our dataset."
 - "The **Disagreement Analysis** (Lines 1103–1111) is the 'Smoking Gun'. It lists all reviews that VADER thought were happy but RoBERTa correctly identified as unhappy."

- **Feature Walkthrough:**

- "The final page is an **Integrated Executive Summary**."
 - "The **Word Clouds** (Lines 1196–1223) show the difference between positive praise and negative complaints at a glance."
 - "The **Age & Department Analysis** (Lines 1184–1191) helps identify if younger or older customers are more unhappy, or if specific departments like 'Dresses' have a quality issue."
 - "Finally, the **Export Section** (Lines 1268–1289) allows stakeholders to download all analyzed data, including irony probabilities, for use in their own business intelligence tools."
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Part 4: Conclusion & Research Findings (17:00 – 20:00)

Action: Show the Heatmap or the Analytics Dashboard again

Say this:

"To conclude my demonstration, I'd like to summarize the key findings and contributions of this research.

4.1 Key Findings

1. **The Sarcasm Blind Spot:** My research found that standard sentiment tools misclassify roughly **3.5% of reviews** due to sarcasm. By using a hybrid RoBERTa + Lexical approach, we recovered these 'lost' negative signals.
2. **Dissatisfaction vs. Rating:** We confirmed that star ratings are often 'sticky'—customers give 3 or 4 stars but express 80/100 dissatisfaction in the text. Our NLP model is more sensitive to these operational issues than the rating column.
3. **Topic Severity:** Unsupervised modeling revealed that **Product Quality** and **Fit/Size** are the primary drivers of severe dissatisfaction in this domain, accounting for over 60% of high-dissatisfaction scores.

4.2 Research Contributions

This project contributes three novel elements:

- **A Continuous Metric:** Moving from binary labels to a 0–100 scale for business-level quantification.
- **Hybrid Sarcasm Architecture:** Combining transformer deep learning with domain-specific lexical heuristics.

- **Operational Heatmaps:** Cross-referencing unsupervised topics with dissatisfaction scores to create a visual 'pain-point' map.

4.3 Limitations and Future Work

While successful, there are limitations:

- **Domain Specificity:** The current model is tuned for e-commerce. Future work could adapt this to the hospitality or healthcare sectors.
- **ABSA Integration:** The next step would be **Aspect-Based Sentiment Analysis**, allowing us to see specifically that 'Zippers' or 'Buttons' are the issue within the Quality topic.

This project demonstrates that quantifying customer dissatisfaction through a hybrid NLP approach provides deeper, more actionable intelligence than traditional methods. Thank you for your time."